

Install and Configure Git on CentOS 7

This article explains to beginners and intermediate Linux users how to install, configure and use Git. It's also for those who are comfortable with Linux but are hesitant to use Git, due to their lack of familiarity with it.



Git is a distributed version control system (DVCS) created by Linus Torvalds. It's mostly used by developers, but it can also be used to store your dot files (a dot file begins with a '.' and the term is generally used to refer to `.bashrc`, `.vimrc` or other such set-up/configuration files), important scripts, etc.

It's different from other similar DVCS systems because it does not treat data as a set of files, but rather, when you save your project in Git, it takes a snapshot of how the files look at that instant in time and stores a reference to it. The core of Git is a key-value data system. No matter what data you insert into Git, it will checksum it using an *SHA 1* algorithm and create a 40-character hex key. Therefore, the name of the file is not really relevant to Git.

GitHub is a Web-based hosting service; it offers a remote Git repository where you can host your files.



Note: You don't need GitHub to use Git. You can host your remote repository on a company server as well.

Git terminology

Let's get familiar with some terminology:

- **Working tree:** This is the directory in which you put the files that you want Git to manage. This is where you store files so that they can be staged to be pushed into the repository. Basically, this could be any directory on your local file system that has a Git repository associated with it. When you run the `git init` command in a normal directory, it transforms the normal directory into a working tree (working directory), because running that command creates a `.git` sub-directory right inside the directory in which you have run the command. We'll soon see how this is done.
- **Repository (Git directory):** The repository, also called the Git directory, is where all your game-changing Git files are stored and where all the Git goodness starts. It basically stores all the meta data, objects, etc, for your project.
- A local repository exists on your laptop and is associated with your working tree. It's the `.git` directory that gets created when you run `git init` as stated earlier.